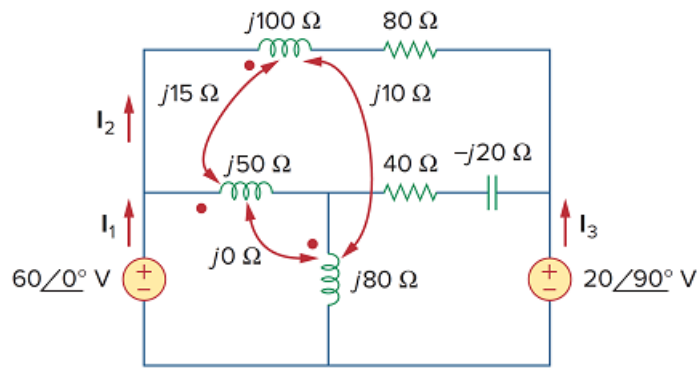


Problem

Use *PSpice* or *MultiSim* to find I_1 , I_2 , and I_3 in the circuit of Fig. 13.141.

Figure 13.141



Step-by-step solution

Step 1 of 2

The schematic is shown below.

$$K_1 = \frac{15}{\sqrt{5000}} = 0.2121 ,$$

$$K_2 = \frac{10}{\sqrt{8000}} = 0.118$$

In the AC sweep box, we type total pts = 1, start frequency = 0.1592 and end frequency = 0.1592. After the circuit is saved and simulated, the output includes,

Frequency	1 M(V_Print1)	1 P(V_Print1)
1.592 E-01	4.068 E-01	-7.786 E+01
Frequency	1 M(V_Print2)	1 P(V_Print2)
1.592 E-01	1.306 E+00	-6.801 E+01
Frequency	1 M(V_Print3)	1 P(V_Print3)
1.592 E-01	1.336 E+00	-5.492 E+01

Thus,

$$I_1 = 1.306 \angle -68.01^\circ \text{ A} ,$$

$$I_2 = 406.8 \angle -77.86^\circ \text{ A} ,$$

$$I_3 = 1.336 \angle -54.92^\circ \text{ A} .$$

Step 2 of 2

